A transitive tail beyond a singleton endpoint

Human Compression project: bounded L044 proof component

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$ be a faithful transitive string C-group, $n \ge 12$, with every maximal omission intransitive. Suppose G_0 has a singleton orbit $\{a\}$, its other orbit contains b, and ab is the unique eligible 0-edge. Assume $K = G_{>1}$ is transitive on $X = \Omega \setminus \{a, b\}$ and fixes a, b.

Proposition 1. *Under these hypotheses,* $r \le \lfloor (n-1)/2 \rfloor$.

Proof. The crossing involution has the form $\rho_0 = (a\ b)\lambda$, with λ an odd involution on X. It centralizes the transitive action of K on X. Therefore λ has no fixed point: a fixed point of a centralizing permutation would propagate to the entire orbit. Write $|X| = 2m$. Then λ consists of m transpositions, so m is odd. As $n = 2m + 2 \ge 12$, one has $m \ge 5$.

The pairs of λ are a K-invariant block system. Every distinguished K generator is essential in the block action. Indeed choose a minimal distinguished subset producing the whole block image. It has at most two point orbits on X; adjoining ρ_0, whose restriction is λ, joins them. If a K generator were omitted by this subset, the corresponding global omission would be transitive on X. That omitted generator is supported on X, so adjoining it cannot merge omission orbits. Since the result is transitive G, the omission would already be transitive, a contradiction. This is the full actual-omission argument of the accepted commuting-tail lemma, specialized to the present pair system.

Whiston's bound now gives $r - 2 \le m - 1$, hence $r \le m + 1$. As $\lfloor (n-1)/2 \rfloor = m$, a failure of the desired bound would force $r = m + 1$ and make the $m - 1$ independent block images generate the full S_m, by Whiston's equality clause.

Put $\sigma = \rho_1$. It fixes a, because it belongs to G_0, and it must move b, because K fixes b whereas $G_0 = \langle \sigma, K \rangle$ is transitive on $\{b\} \cup X$. Thus for some $x \in X$ one has

$$\sigma = (b\ x)v,$$

where v is an odd involution supported on $X \setminus \{x\}$. Let $x' = x^\lambda$ and $H = G_{>2}$. All its generators fix b and commute with σ, so fix x. They also commute with λ, so fix x'. Their $m - 2$ block images are independent and fix the pair $\{x, x'\}$; Whiston makes their image the full S_{m-1} on the other pairs.

The only H-fixed points in X are x, x', since its block image moves every other pair. As v centralizes H and fixes x, it fixes x' as well. On the remaining $m - 1$ pairs, the accepted paired-row centralizer lemma applies: their number is even and at least four, and the induced group is S_{m-1}. Hence v is even on the remaining points. This contradicts the oddness of v, excludes $r = m + 1$ and proves the proposition. □

Source and scope. This is a complete replacement proof of the transitive-tail singleton-endpoint case of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.29, physical page 38, in the main $n \ge 12$ range. It reuses the accepted commuting-tail and paired-row lemmas and retained Whiston bounds. The tight case is excluded without importing the unproved full equality-graph assertion of source Proposition 5.21. No finite computation, projected IP or new formal theorem is required. The intransitive tail, the distinct D_{10} residual, later table exhaustions and the complete L044 source remain separate. No original or question credit is asserted.